

Data Warehouse Modeling

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Data Analytics

- Overview
- Data Warehousing
- Online Analytical Processing
- Data Mining

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Overview

- **Data analytics:** the processing of data to infer patterns, correlations, or models for prediction
- Primarily used to make business decisions
 - Per individual customer
 - E.g., what product to suggest for purchase
 - Across all customers
 - E.g., what products to manufacture/stock, in what quantity
- Critical for businesses today



Overview (Cont.)

- Common steps in data analytics
 - Gather data from multiple sources into one location
 - Data warehouses also integrated data into common schema
 - Data often needs to be **extracted** from source formats, **transformed** to common schema, and **loaded** into the data warehouse
 - Can be done as **ETL (extract-transform-load)**, or **ELT (extract-load-transform)**
 - Generate aggregates and reports summarizing data
 - Dashboards showing graphical charts/reports
 - **Online analytical processing (OLAP) systems** allow interactive querying
 - Statistical analysis using tools such as R/SAS/SPSS
 - Including extensions for parallel processing of big data
 - Build **predictive models** and use the models for decision making



Overview (Cont.)

- Predictive models are widely used today
 - E.g., use customer profile features (e.g. income, age, gender, education, employment) and past history of a customer to predict likelihood of default on loan
 - and use prediction to make loan decision
 - E.g., use past history of sales (by season) to predict future sales
 - And use it to decide what/how much to produce/stock
 - And to target customers
- Other examples of business decisions:
 - What items to stock?
 - What insurance premium to change?
 - To whom to send advertisements?



Overview (Cont.)

- **Machine learning** techniques are key to finding patterns in data and making predictions
- **Data mining** extends techniques developed by machine-learning communities to run them on very large datasets
- The term **business intelligence (BI)** is synonym for data analytics
- The term **decision support** focuses on reporting and aggregation

The Evolution

- 1960 - DSS processing using Fortran or COBOL
- 1970 - DBMS systems and the advent of DASD
- 1975 - OLTP systems facilitating faster access to data
- 1980 - PC/4GL technology and the advent of MIS
- 1985 - OLAP systems and separation of analytical processing from transactional processing
- 1994 - Architected environments with integrated OLAP engines and tools

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What is Data Warehouse ?

- A copy of transaction data specifically structured to Query and Analysis (Ralph Kimball, 1996)
- A collection of integrated, subject oriented databases designed to support the DSS function where each unit of data is relevant at some moment of time (Bill Inmon, 1991)
- The data characteristics of a Data Warehouse are;
 - Subject-oriented
 - Time-variant
 - Non-volatile
 - Integrated

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What is Data Warehouse ? ... (cont'd)

- A single, complete and consistent store of data obtained from a variety of different sources made available to end users, in what they can understand and use in a business context (Barry Devlin 1992)
- A process of transforming data into information and making it available to users in a timely enough manner to make a difference (Forrester Research 1996)

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Data Warehouse Goals/Characteristics

- It must make an organization's information easily accessible (slicing and dicing)
- It must present the organization's information consistently
- It must be adaptive and resilient to change
- It must be a secure bastion that protects our information assets
- It must serve as the foundation for improved decision making
- The business community must accept the DW, if it is to be deemed successful

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Data Warehouse Concepts

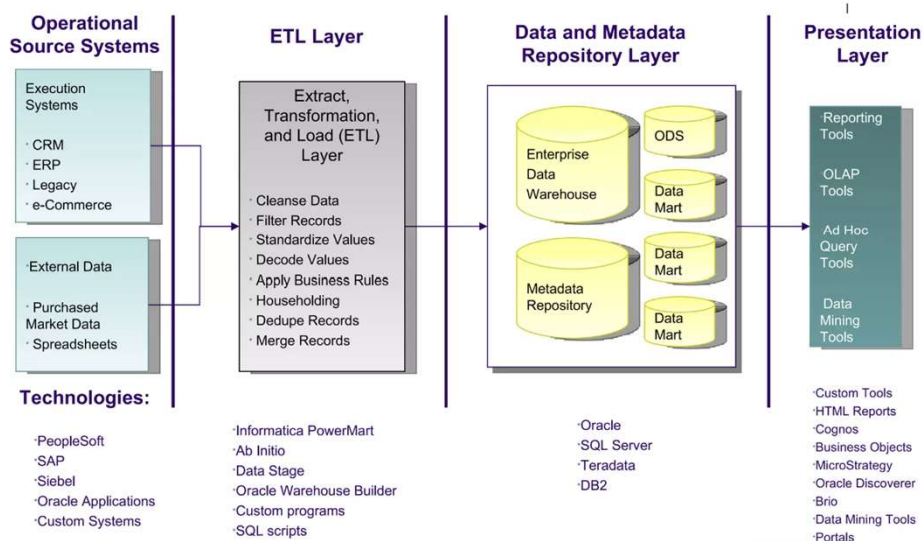
Data Warehouse (OLAP) and OLTP

Characteristics	On-Line Transaction Processing (OLTP)	Data Warehouse
Data Content	Current values	Historical data, summarized data, calculated data
Data Organization	Application by application	Subject areas across enterprise
Nature of Data	Dynamic	Static until refreshed, based on frequency
Data Manipulation	Updated on a field-by-field basis	Accessed & manipulated usually no direct update
Usage	Highly structured, repetitive processing (Clerical User)	Highly structured, analytical processing (Knowledge User)
Response Time	Critical (Sub-Second to several seconds)	Several seconds to minutes
Updates vs. Reports	Real-time Updates, Batch Reporting	Batch Updates, Real-time Reporting

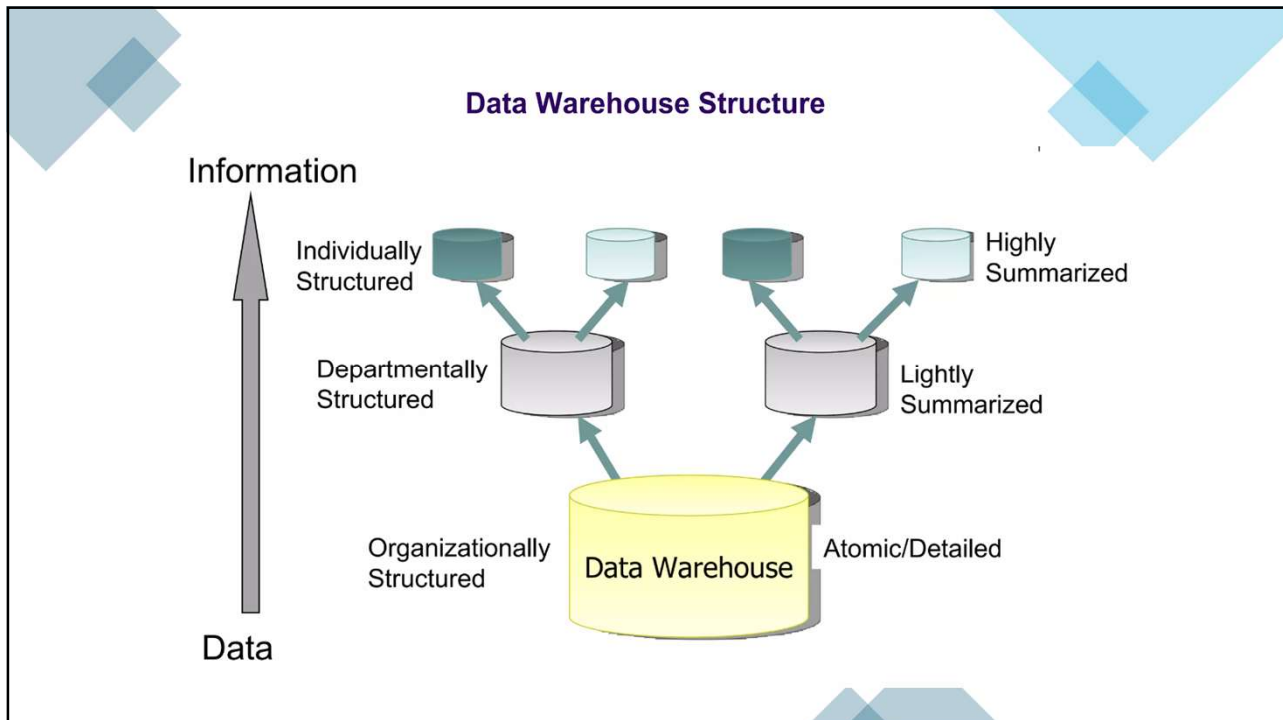
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Data Warehouse Architecture



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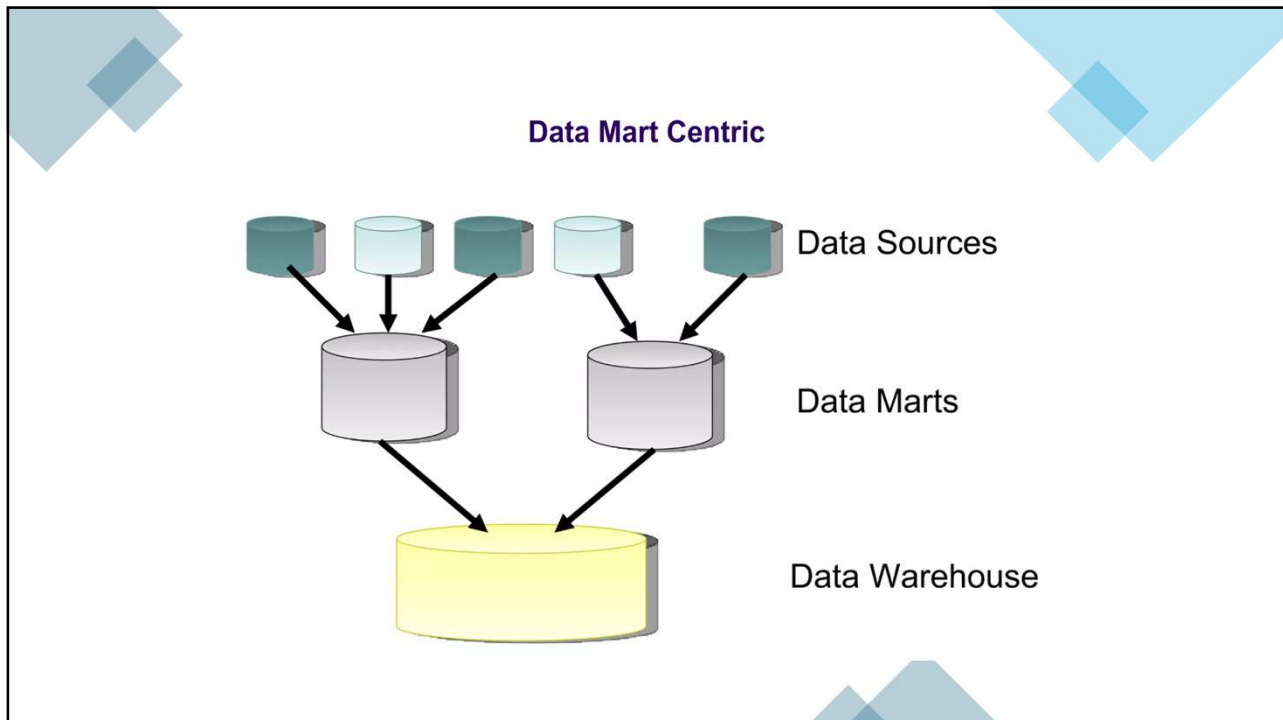
Data Warehouse Architecture Drivers

The requirements that drive the DW architecture are;

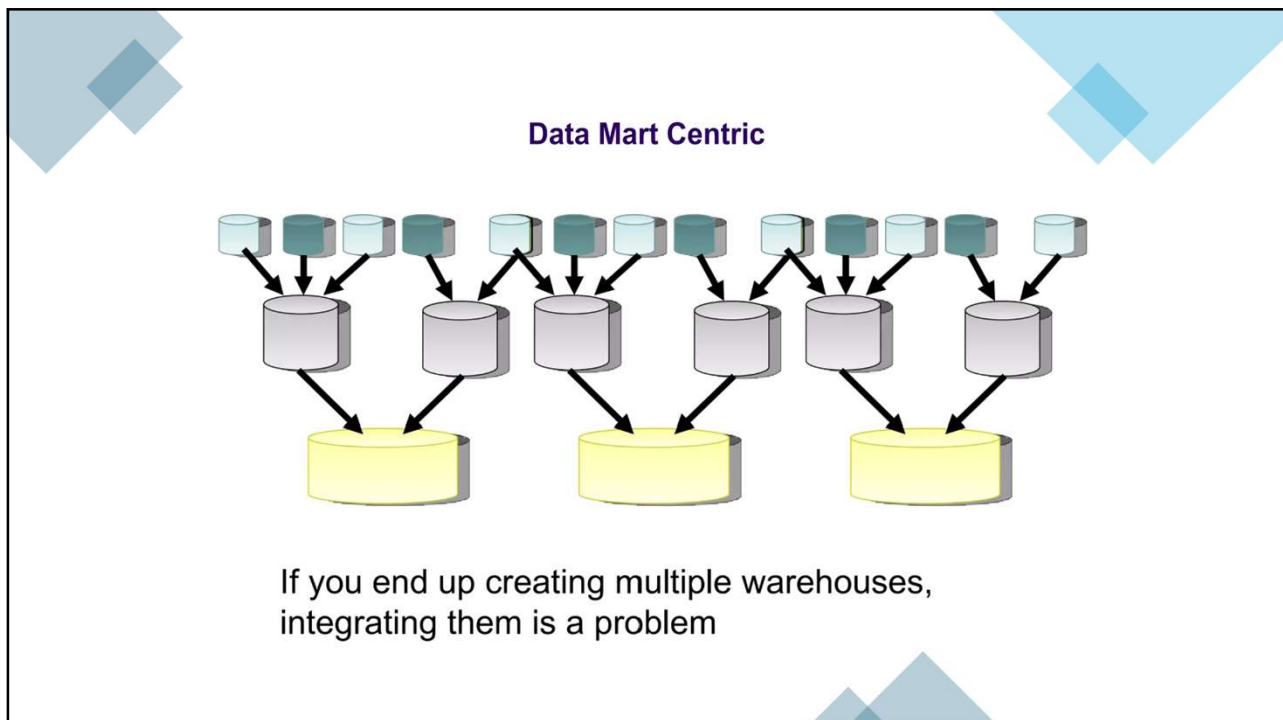
- Granularity of data
- Data retention and timeliness
- Reporting capability
- Availability
- Scalability

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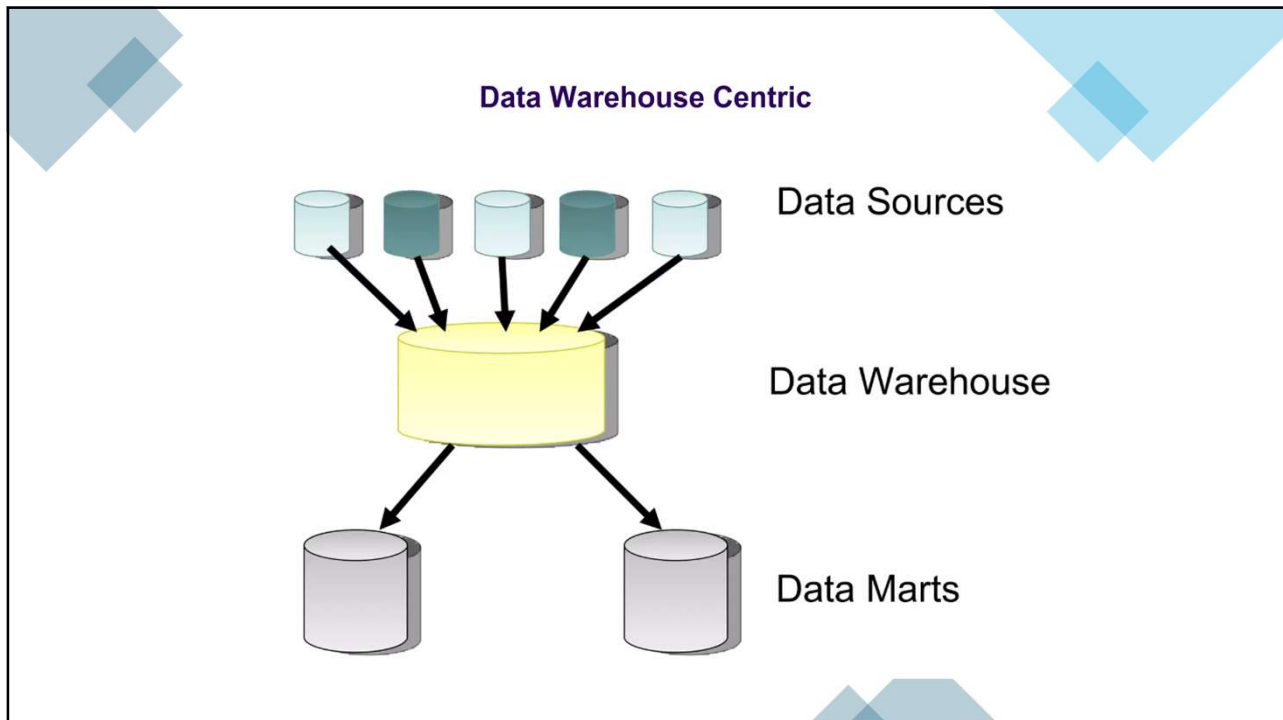
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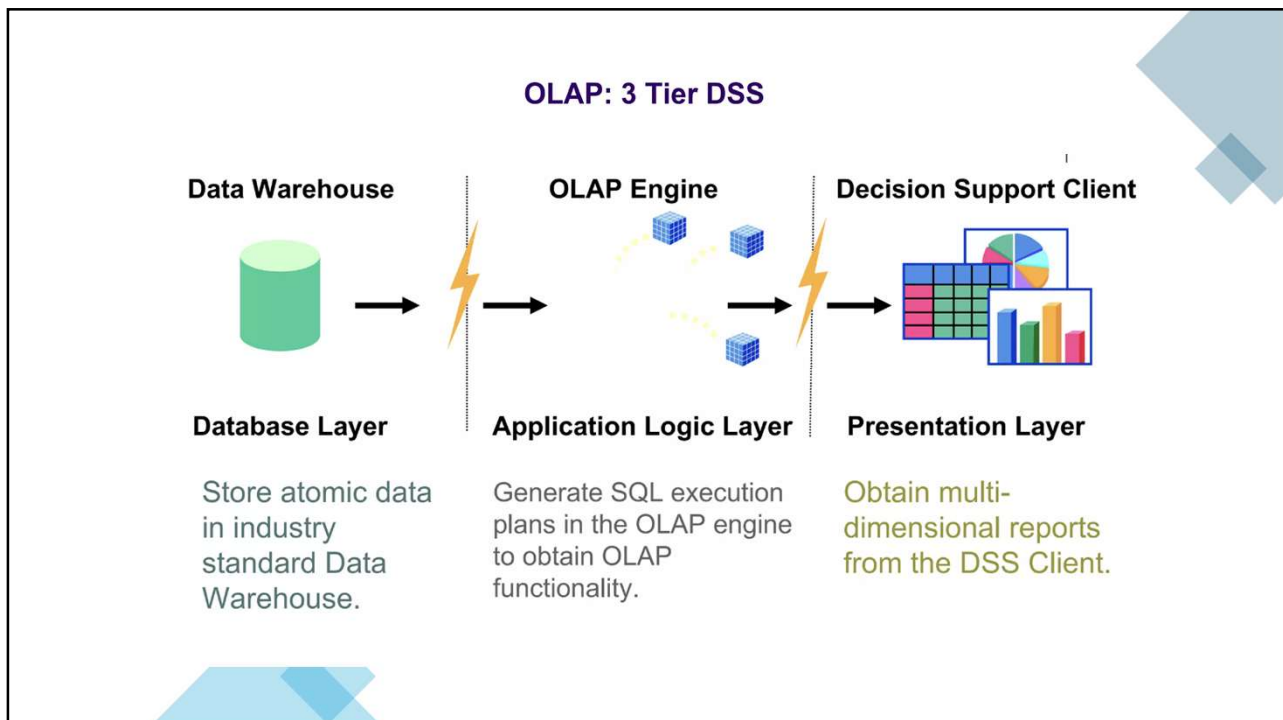
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OLAP Servers

- Support multidimensional OLAP queries
- Characterized by how the underlying data is stored
- Multidimensional OLAP (MOLAP) Servers
 - Data stored in array based structures e.g. Hyperion Essbase
- Relational OLAP (ROLAP) Servers
 - Data stored in relational tables e.g. Microstrategy, IBM Informix
- Hybrid OLAP (HOLAP) Servers
 - Data distributed between relational and specialized storage e.g. Cognos, Microsoft Analysis Services

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OLAP Operations

- Rollup; summarize operations
 - E.g. given sales data, summarize sales for last year by product category and region
- Drill down; get more details
 - E.g. given summarized sales as above, find breakup of sales within each region
- Slice and dice; select and project
 - Sales of soft-drinks in Gothenburg over the last quarter
- Pivot; change the view of data

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Strengths OLAP

- It is a powerful visualization tool
- It provides fast, interactive response times
- It is good for analyzing time series
- It can be useful to find some clusters and outliers
- Many vendors offer OLAP tools

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Dimensional Modeling

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What is Dimensional Modeling ?

- Logical design technique that seeks to present the data in a standard, intuitive framework that allows for high-performance access.
- Adheres to a discipline that uses the relational model with some important restrictions.
- Composed of one table with a multi-part key, called the fact table, and a set of smaller tables called dimension tables.

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DM v/s ER Models

DM	ER
Used to design database for Online Analytical Processing (OLAP)	Used to design database for Online Transaction Processing (OLTP)
Support ad hoc end-user queries	Support defined queries
Intuitive & facilitates high-performance retrieval of data	Removes redundancy of data
De-normalized	Normalized

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Fact Tables

- Primary table in the DM
- Each row corresponds to a measurement
- Facts in the fact table are numeric and additive
- Narrow rows with a few columns
- Large number of rows (billions)
- Express many-to-many relationships between dimensions

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Dimension Tables

- Define business in terms already familiar to users
- Implement the user interface to the DW
- Wide rows with lots of descriptive text
- Small tables (about a million rows)
- Joined to fact table by a foreign key
- Heavily indexed
- E.g. of typical dimensions
 - time periods, geographic region (markets, cities), products, customers, salesperson, etc.

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Warehouse Models & Operators

- Data Models
 - relations
 - stars & snowflakes
 - cubes
- Operators
 - slice & dice
 - roll-up, drill down
 - pivoting
 - other

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Multi-Dimensional Data

- Measures - numerical (and additive) data being tracked in business, can be analyzed and examined
- Dimensions - business parameters that define a transaction, relatively static data such as lookup or reference tables
- Example: Analyst may want to view *sales* data (measure) by *geography*, by *time*, and by *product* (dimensions)

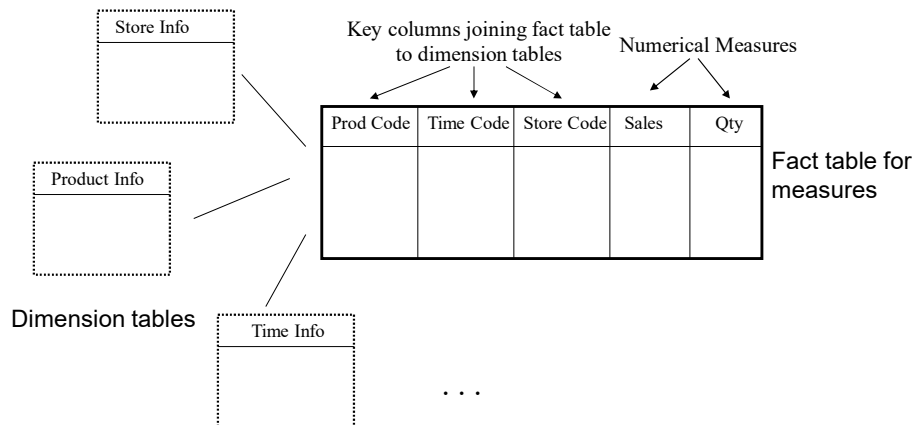
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The Multi-Dimensional Model

“Sales by product line over the past six months”

“Sales by store between 1990 and 1995”



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Multidimensional Modeling

- Multidimensional modeling is a technique for structuring data around the business concepts
- ER models describe “entities” and “relationships”
- Multidimensional models describe “measures” and “dimensions”

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Dimensional Modeling

- Dimensions are organized into hierarchies
 - E.g., Time dimension: days → weeks → quarters
 - E.g., Product dimension: product → product line → brand
- Dimensions have attributes

Time

Date
Month
Year

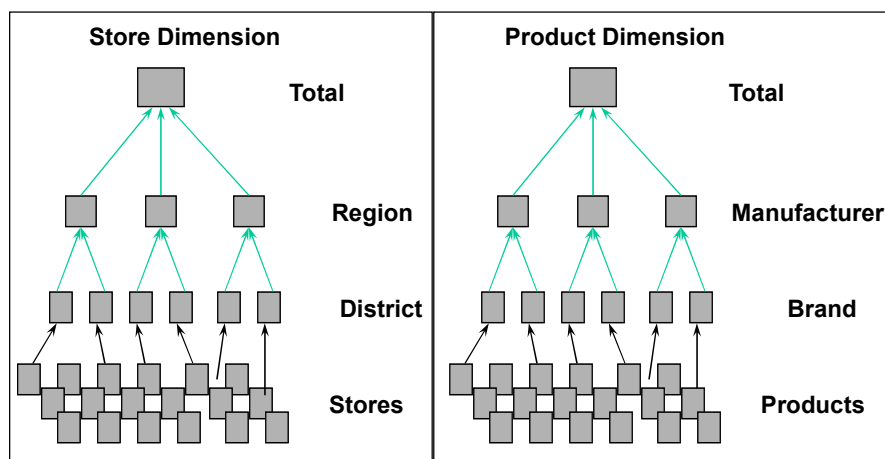
Store

StoreID
City
State
Country
Region

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Dimension Hierarchies



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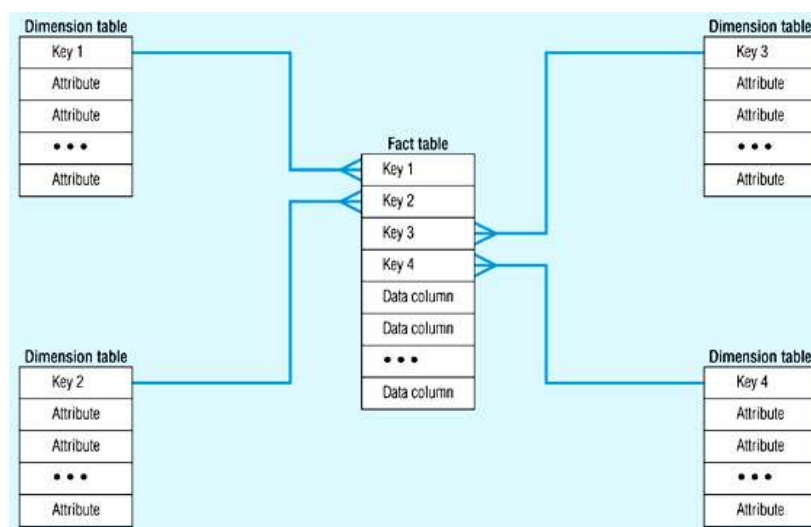
Schema Design

- Most data warehouses use a star schema to represent multi-dimensional model.
- Each dimension is represented by a **dimension table** that describes it.
- A **fact table** connects to all dimension tables with a multiple join. Each tuple in the fact table consists of a pointer to each of the dimension tables that provide its multi-dimensional coordinates and stores measures for those coordinates.
- The links between the fact table in the center and the dimension tables in the extremities form a shape like a star.

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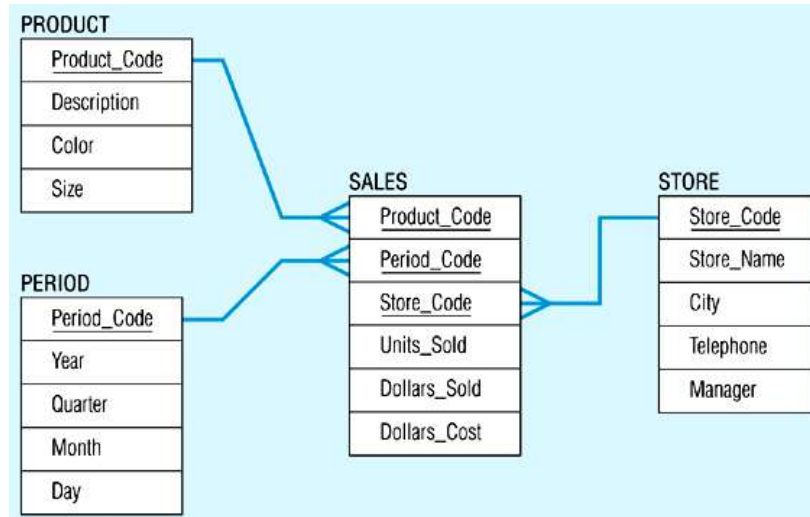
Star Schema (in RDBMS)



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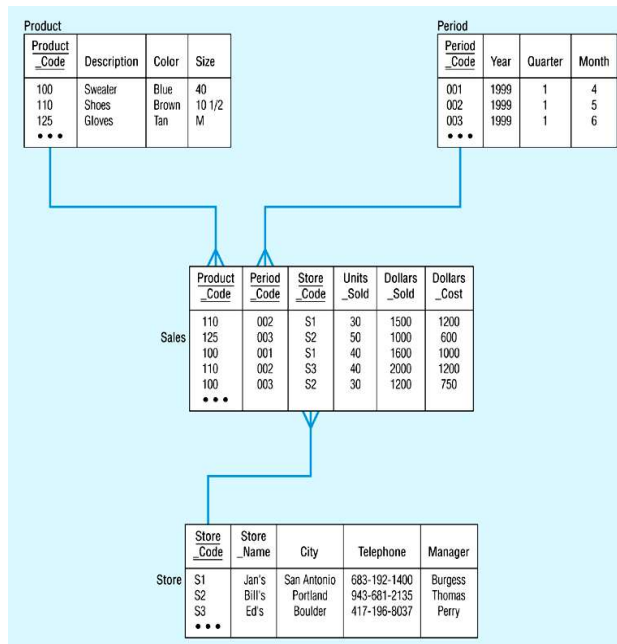
Star Schema Example



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Star Schema with Sample Data



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The “Classic” Star Schema

- ♦ A relational model with a one-to-many relationship between dimension table and fact table.
- ♦ A single fact table, with detail and summary data
- ♦ Fact table primary key has only one key column per dimension
- ♦ Each dimension is a single table, highly denormalized
- **Benefits:** Easy to understand, intuitive mapping between the business entities, easy to define hierarchies, reduces # of physical joins, low maintenance, very simple metadata
- **Drawbacks:** Summary data in the fact table yields poorer performance for summary levels, huge dimension tables a problem

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Need for Aggregates

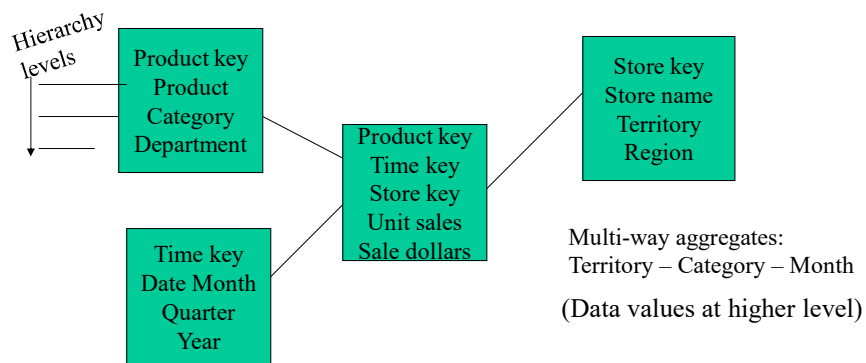
- Sizes of typical tables:
 - Time dimension: 5 years x 365 days = 1825
 - Store dimension: 300 stores reporting daily sales
 - Production dimension: 40,000 products in each store (about 4000 sell in each store daily)
 - Maximum number of base fact table records: 2 billion (lowest level of detail)
- A query involving 1 brand, all store, 1 year: retrieve/summarize over 7 million fact table rows.

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Aggregating Fact Tables

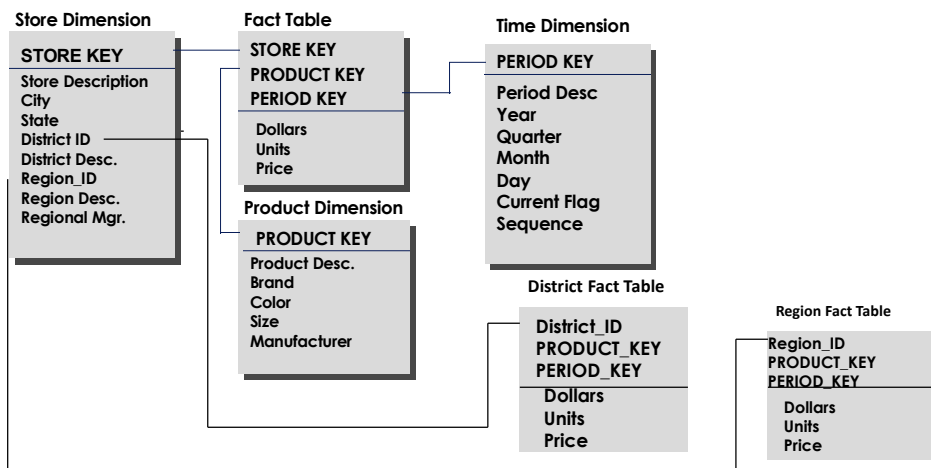
- Aggregate fact tables are summaries of the most granular data at higher levels along the dimension hierarchies.



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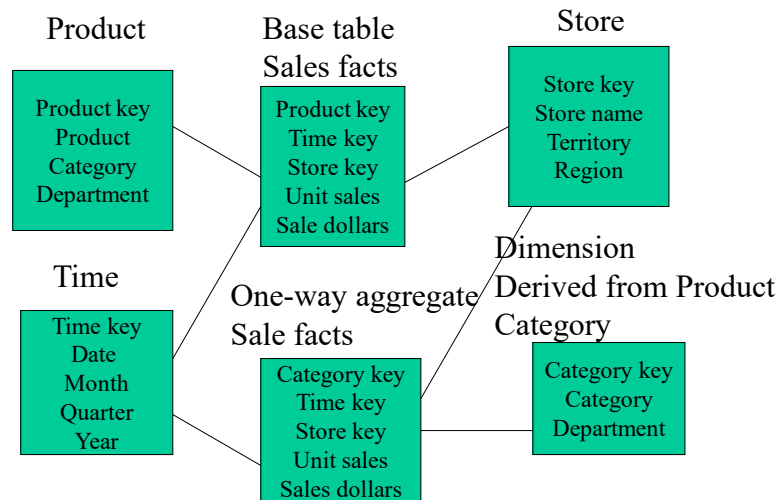
The “Fact Constellation” Schema



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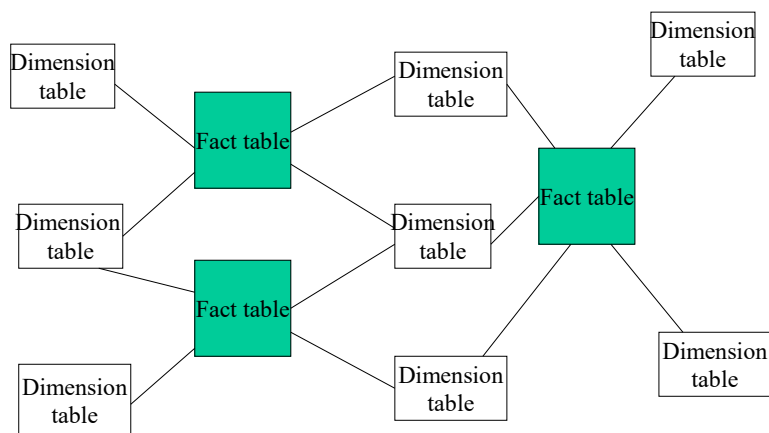
Aggregate Fact Tables



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Families of Stars



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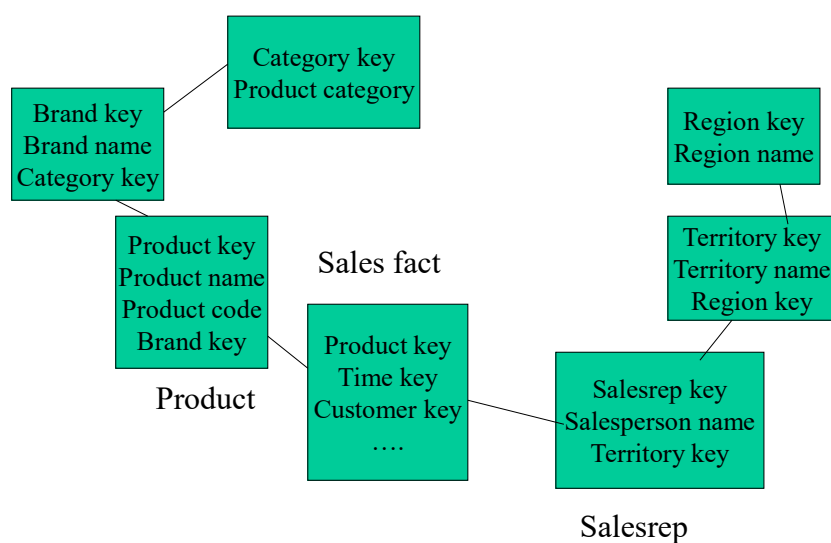
Snowflake Schema

- Snowflake schema is a type of star schema but a more complex model.
- “Snowflaking” is a method of normalizing the dimension tables in a star schema.
- The normalization eliminates redundancy.
- The result is more complex queries and reduced query performance.

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Sales: Snowflake Schema

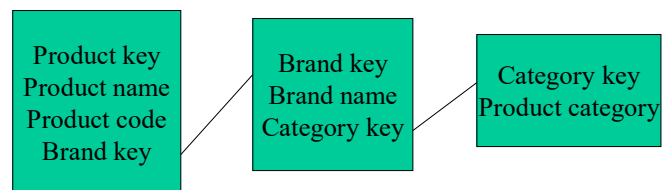


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Snowflaking

- The attributes with low cardinality in each original dimension table are removed to form separate tables. These new tables are linked back to the original dimension table through artificial keys.



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Snowflake Schema

- Advantages:
 - Small saving in storage space
 - Normalized structures are easier to update and maintain
- Disadvantages:
 - Schema less intuitive and end-users are put off by the complexity
 - Ability to browse through the contents difficult
 - Degrade query performance because of additional joins

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What is the Best Design?

- Performance benchmarking can be used to determine what is the best design.
- Snowflake schema: easier to maintain dimension tables when dimension tables are very large (reduce overall space). It is not generally recommended in a data warehouse environment.
- Star schema: more effective for data cube browsing (less joins): can affect performance.

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Aggregates

- Add up amounts for day 1
- In SQL: `SELECT sum(amt) FROM SALE WHERE date = 1`

sale	prodId	storeId	date	amt
	p1	s1	1	12
	p2	s1	1	11
	p1	s3	1	50
	p2	s2	1	8
	p1	s1	2	44
	p1	s2	2	4



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Aggregates

- Add up amounts by day
- In SQL: `SELECT date, sum(amt) FROM SALE GROUP BY date`

sale	prodId	storeId	date	amt
	p1	s1	1	12
	p2	s1	1	11
	p1	s3	1	50
	p2	s2	1	8
	p1	s1	2	44
	p1	s2	2	4



ans	date	sum
	1	81
	2	48

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Another Example

- Add up amounts by day, product
- In SQL: `SELECT date, sum(amt) FROM SALE GROUP BY date, prodId`

sale	prodId	storeId	date	amt
	p1	s1	1	12
	p2	s1	1	11
	p1	s3	1	50
	p2	s2	1	8
	p1	s1	2	44
	p1	s2	2	4



sale	prodId	date	amt
	p1	1	62
	p2	1	19
	p1	2	48

→ rollup →

← drill-down ←

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Aggregates

- Operators: sum, count, max, min, median, ave
- “Having” clause
- Using dimension hierarchy
 - average by region (within store)
 - maximum by month (within date)

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Data Cube

Fact table view:

sale	prodId	storeId	amt
	p1	s1	12
	p2	s1	11
	p1	s3	50
	p2	s2	8

Multi-dimensional cube:

	s1	s2	s3
p1	12		50
p2	11	8	

dimensions = 2

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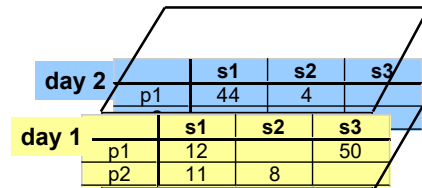
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3-D Cube

Fact table view:

sale	prodid	storeld	date	amt
	p1	s1	1	12
	p2	s1	1	11
	p1	s3	1	50
	p2	s2	1	8
	p1	s1	2	44
	p1	s2	2	4

Multi-dimensional cube:

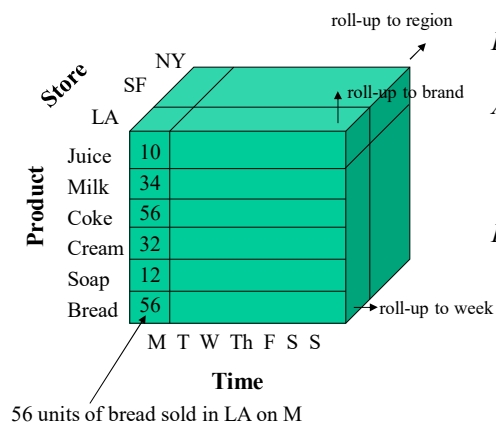


dimensions = 3

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Example



Dimensions:

Time, Product, Store

Attributes:

Product (upc, price, ...)

Store ...

...

Hierarchies:

Product → Brand → ...

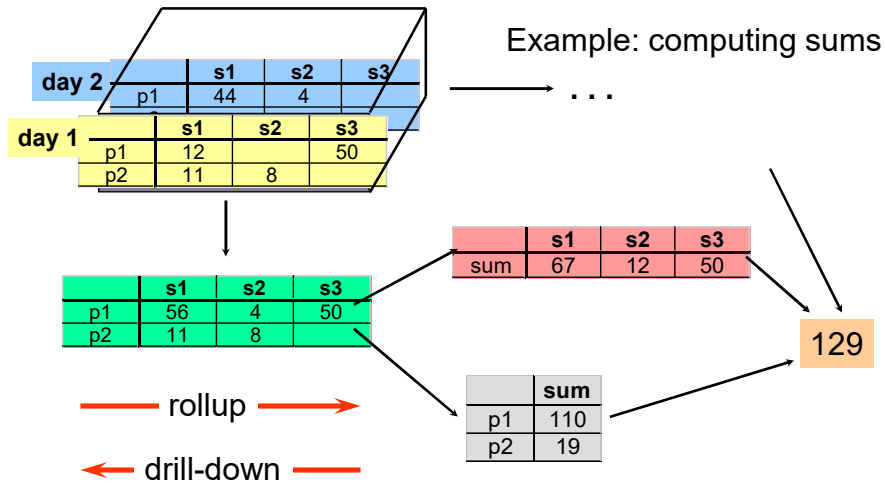
Day → Week → Quarter

Store → Region → Country

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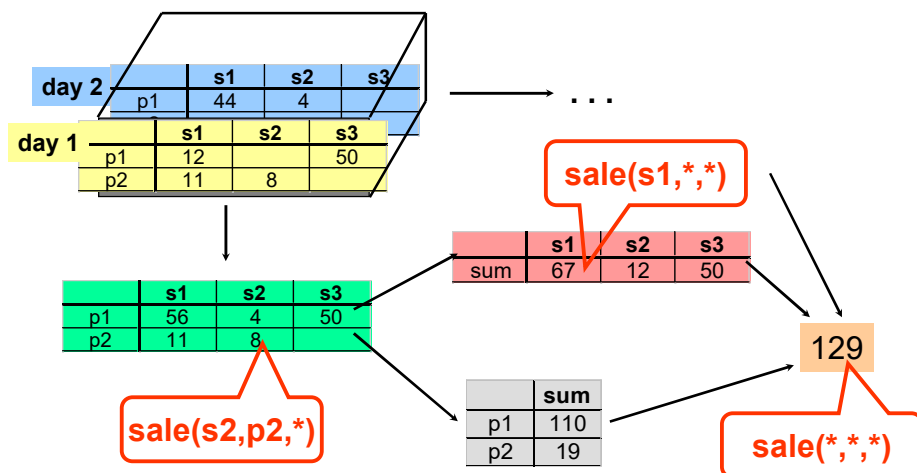
Cube Aggregation: Roll-up



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Cube Operators for Roll-up



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Extended Cube

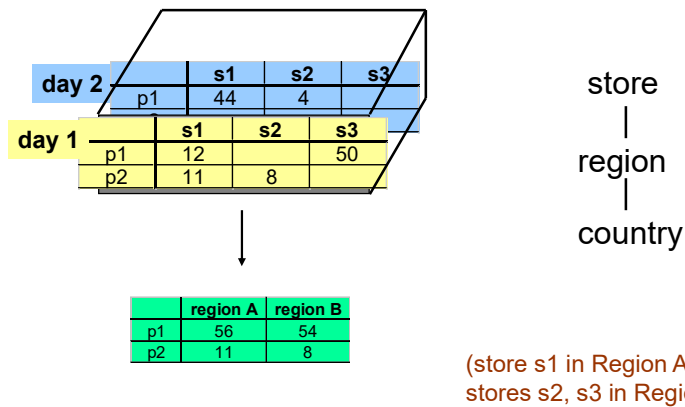
		*	s1	s2	s3	
day 2	p1		56	4	50	110
	p2		11	8		19
	*					129
day 1	p1	44	4			48
	p2	12		50	62	
	*	23	8	50	81	

sale(*,p2,*)

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Aggregation Using Hierarchies



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Slicing

TIME = day 1

	s1	s2	s3
p1	12		50
p2	11	8	

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Slicing & Pivoting

		Sales (\$ millions)		
		Products	Time	
			d1	d2
Store s1	Electronics	\$5.2		
	Toys	\$1.9		
	Clothing	\$2.3		
	Cosmetics	\$1.1		
Store s2	Electronics	\$8.9		
	Toys	\$0.75		
	Clothing	\$4.6		
	Cosmetics	\$1.5		

		Sales (\$ millions)	
		Products	d1
			Store s1
Store s1	Electronics	\$5.2	\$8.9
	Toys	\$1.9	\$0.75
	Clothing	\$2.3	\$4.6
	Cosmetics	\$1.1	\$1.5
Store s2	Electronics		
	Toys		
	Clothing		

An arrow indicates the pivot operation from the first table to the second, showing the transformation of the data layout.

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Summary of Operations

- Aggregation (roll-up)
 - aggregate (summarize) data to the next higher dimension element
 - e.g., total sales by city, year → total sales by region, year
- Navigation to detailed data (drill-down)
- Selection (slice) defines a subcube
 - e.g., sales where city = 'Gainesville' and date = '1/15/90'
- Calculation and ranking
 - e.g., top 3% of cities by average income
- Visualization operations (e.g., Pivot)
- Time functions
 - e.g., time average

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Query & Analysis Tools

- Query Building
- Report Writers (comparisons, growth, graphs,...)
- Spreadsheet Systems
- Web Interfaces
- Data Mining

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OLAP

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Data Analysis and OLAP

- **Online Analytical Processing (OLAP)**
 - Interactive analysis of data, allowing data to be summarized and viewed in different ways in an online fashion (with negligible delay)
- We use the following relation to illustrate OLAP concepts
 - *sales* (*item_name*, *color*, *clothes_size*, *quantity*)

This is a simplified version of the *sales* fact table joined with the dimension tables, and many attributes removed (and some renamed)

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Example sales relation

item_name	color	clothes_size	quantity
dress	dark	small	2
dress	dark	medium	6
dress	dark	large	12
dress	pastel	small	4
dress	pastel	medium	3
dress	pastel	large	3
dress	white	small	2
dress	white	medium	3
dress	white	large	0
pants	dark	small	14
pants	dark	medium	6
pants	dark	large	0
pants	pastel	small	1
pants	pastel	medium	0
pants	pastel	large	1
pants	white	small	3
pants	white	medium	0
pants	white	large	2
shirt	dark	small	2
shirt	dark	medium	6
shirt	dark	large	6
shirt	pastel	small	4
shirt	pastel	medium	1
shirt	pastel	large	2
shirt	white	small	17
shirt	white	medium	1
shirt	white	large	10
skirt	dark	small	2
skirt	dark	medium	5
...	...	...	...
...	...	...	...

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Cross Tabulation of sales by *item_name* and *color*

clothes_size **all**

	<i>color</i>			
	dark	pastel	white	total
<i>item_name</i> skirt	8	35	10	53
dress	20	10	5	35
shirt	14	7	28	49
pants	20	2	5	27
total	62	54	48	164

- The table above is an example of a **cross-tabulation (cross-tab)**, also referred to as a **pivot-table**.
 - Values for one of the dimension attributes form the row headers
 - Values for another dimension attribute form the column headers
 - Other dimension attributes are listed on top
 - Values in individual cells are (aggregates of) the values of the dimension attributes that specify the cell.

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Data Cube

- A **data cube** is a multidimensional generalization of a cross-tab
- Can have n dimensions; we show 3 below
- Cross-tabs can be used as views on a data cube

		item_name					clothes_size			
		skirt	dress	shirt	pants	all	small	medium	large	all
color	dark	8	20	14	20	62	4	16	34	16
	pastel	35	10	7	2	54	9	18	21	18
	white	10	5	28	5	48	42	45	77	45
	all	53	35	49	27	164	77	108	154	239

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Online Analytical Processing Operations

- **Pivoting**: changing the dimensions used in a cross-tab
 - E.g., moving colors to column names
- **Slicing**: creating a cross-tab for fixed values only
 - E.g., fixing color to white and size to small
 - Sometimes called **dicing**, particularly when values for multiple dimensions are fixed.
- **Rollup**: moving from finer-granularity data to a coarser granularity
 - E.g., aggregating away an attribute
 - E.g., moving from aggregates by day to aggregates by month or year
- **Drill down**: The opposite operation - that of moving from coarser-granularity data to finer-granularity data

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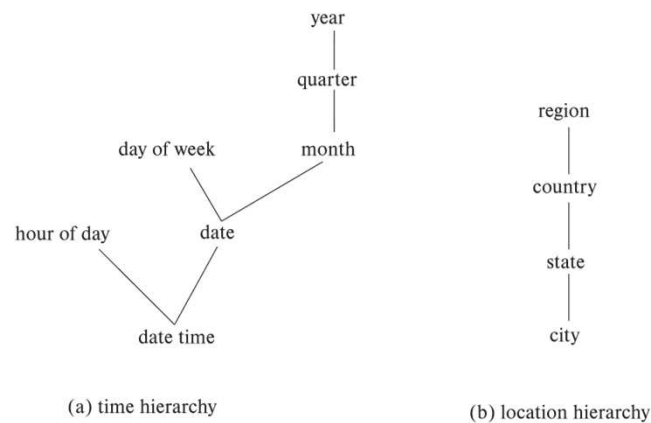
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Hierarchies on Dimensions

- **Hierarchy** on dimension attributes: lets dimensions be viewed at different levels of detail
- E.g., the dimension *datetime* can be used to aggregate by hour of day, date, day of week, month, quarter or year



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Cross Tabulation With Hierarchy

- Cross-tabs can be easily extended to deal with hierarchies
- Can drill down or roll up on a hierarchy
- E.g. hierarchy: *item_name* → *category*

clothes_size: all

category	item_name	color			total	
		dark	pastel	white		
womenswear	skirt	8	8	10	53	88
	dress	20	20	5	35	
	subtotal	28	28	15		
menswear	pants	14	14	28	49	76
	shirt	20	20	5	27	
	subtotal	34	34	33		
total		62	62	48		164

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Relational Representation of Cross-tabs

- Cross-tabs can be represented as relations
- We use the value **all** to represent aggregates.
- The SQL standard actually uses *null* values in place of **all**
 - Works with any data type
 - But can cause confusion with regular null values.

<i>item_name</i>	<i>color</i>	<i>clothes_size</i>	<i>quantity</i>
skirt	dark	all	8
skirt	pastel	all	35
skirt	white	all	10
skirt	all	all	53
dress	dark	all	20
dress	pastel	all	10
dress	white	all	5
dress	all	all	35
shirt	dark	all	14
shirt	pastel	all	7
shirt	white	all	28
shirt	all	all	49
pants	dark	all	20
pants	pastel	all	2
pants	white	all	5
pants	all	all	27
all	dark	all	62
all	pastel	all	54
all	white	all	48
all	all	all	164



OLAP IN SQL



Pivot Operation

- **select** *
from *sales*
pivot (
 sum(*quantity*)
 for *color* **in** ('dark','pastel','white')
)
order by *item name*;

<i>item_name</i>	<i>clothes_size</i>	<i>dark</i>	<i>pastel</i>	<i>white</i>
dress	small	2	4	2
dress	medium	6	3	3
dress	large	12	3	0
pants	small	14	1	3
pants	medium	6	0	0
pants	large	0	1	2
shirt	small	2	4	17
shirt	medium	6	1	1
shirt	large	6	2	10
skirt	small	2	11	2
skirt	medium	5	9	5
skirt	large	1	15	3



Cube Operation

- The **cube** operation computes union of **group by**'s on every subset of the specified attributes
- E.g., consider the query
 select *item_name, color, size, sum(number)*
 from *sales*
 group by cube(*item_name, color, size*)

This computes the union of eight different groupings of the *sales* relation:

{ (*item_name, color, size*), (*item_name, color*),
 (*item_name, size*), (*color, size*),
 (*item_name*), (*color*),
 (*size*), () }

where () denotes an empty **group by** list.

- For each grouping, the result contains the null value for attributes not present in the grouping.



Online Analytical Processing Operations

- Relational representation of cross-tab that we saw earlier, but with *null* in place of **all**, can be computed by

```
select item_name, color, sum(number)
from sales
group by cube(item_name, color)
```

- The function **grouping()** can be applied on an attribute
 - Returns 1 if the value is a null value representing all, and returns 0 in all other cases.

```
select case when grouping(item_name) = 1 then 'all'
           else item_name end as item_name,
       case when grouping(color) = 1 then 'all'
           else color end as color,
       'all' as clothes size, sum(quantity) as quantity
from sales
group by cube(item name, color);
```



Online Analytical Processing Operations

- Can use the function **decode()** in the **select** clause to replace such nulls by a value such as **all**
 - E.g., replace *item_name* in first query by
decode(grouping(item_name), 1, 'all', item_name)



Extended Aggregation (Cont.)

- The **rollup** construct generates union on every prefix of specified list of attributes
- ```
select item_name, color, size, sum(number)
from sales
group by rollup(item_name, color, size)
```

Generates union of four groupings:

$$\{ (item\_name, color, size), (item\_name, color), (item\_name), ( ) \}$$
- Rollup can be used to generate aggregates at multiple levels of a hierarchy.
- E.g., suppose table *itemcategory*(*item\_name*, *category*) gives the category of each item. Then
 

```
select category, item_name, sum(number)
from sales, itemcategory
where sales.item_name = itemcategory.item_name
group by rollup(category, item_name)
```

would give a hierarchical summary by *item\_name* and by *category*.



## Extended Aggregation (Cont.)

- Multiple rollups and cubes can be used in a single group by clause
  - Each generates set of group by lists, cross product of sets gives overall set of group by lists
- E.g.,
 

```
select item_name, color, size, sum(number)
from sales
group by rollup(item_name), rollup(color, size)
```

generates the groupings

$$\{item\_name, ()\} \times \{(color, size), (color), ()\}$$

$$= \{ (item\_name, color, size), (item\_name, color), (item\_name), (color, size), (color), () \}$$
- ```
select item_name, color, clothes_size, sum(quantity)
from sales
group by grouping sets ((color, clothes_size),
(clothes_size, item_name));
```



OLAP Implementation

- The earliest OLAP systems used multidimensional arrays in memory to store data cubes, and are referred to as **multidimensional OLAP (MOLAP)** systems.
- OLAP implementations using only relational database features are called **relational OLAP (ROLAP)** systems
- Hybrid systems, which store some summaries in memory and store the base data and other summaries in a relational database, are called **hybrid OLAP (HOLAP)** systems.



OLAP Implementation (Cont.)

- Early OLAP systems precomputed *all* possible aggregates in order to provide online response
 - Space and time requirements for doing so can be very high
 - 2^n combinations of **group by**
 - It suffices to precompute some aggregates, and compute others on demand from one of the precomputed aggregates
 - Can compute aggregate on *(item_name, color)* from an aggregate on *(item_name, color, size)*
 - For all but a few “non-decomposable” aggregates such as *median*
 - is cheaper than computing it from scratch
- Several optimizations available for computing multiple aggregates
 - Can compute aggregate on *(item_name, color)* from an aggregate on *(item_name, color, size)*
 - Can compute aggregates on *(item_name, color, size)*, *(item_name, color)* and *(item_name)* using a single sorting of the base data



Reporting and Visualization

- **Reporting tools** help create formatted reports with tabular/graphical representation of data
 - E.g., SQL Server reporting services, Crystal Reports
- **Data visualization** tools help create interactive visualization of data
 - E.g., Tableau, FusionChart, plotly, Datawrapper, Google Charts, etc.
 - Frontend typically based on HTML+JavaScript

Acme Supply Company, Inc.
Quarterly Sales Report

Period: Jan. 1 to March 31, 2009

Region	Category	Sales	Subtotal
North	Computer Hardware	1,000,000	1,500,000
	Computer Software	500,000	
	All categories		
South	Computer Hardware	200,000	600,000
	Computer Software	400,000	
	All categories		
Total Sales			2,100,000